

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
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S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimeter-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 903/2

Bridge Chainage – 897+372

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – ROB

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

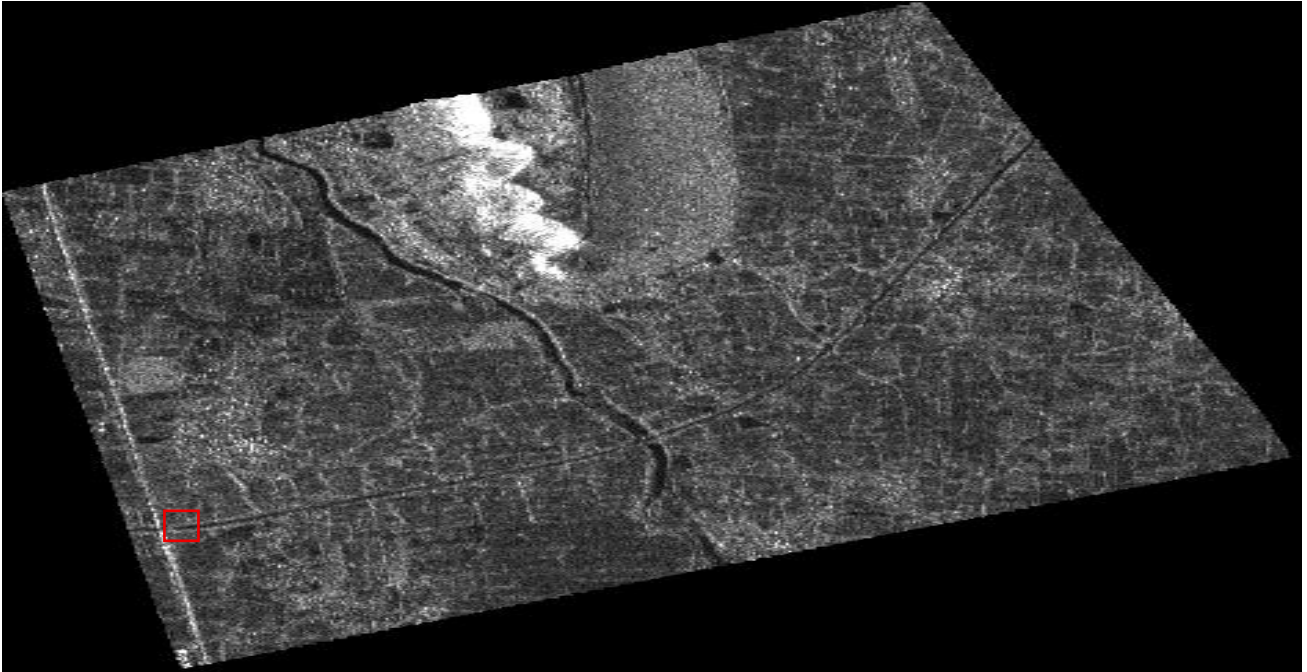
Flight Direction - DESCENDING

Polarization – VV+VH

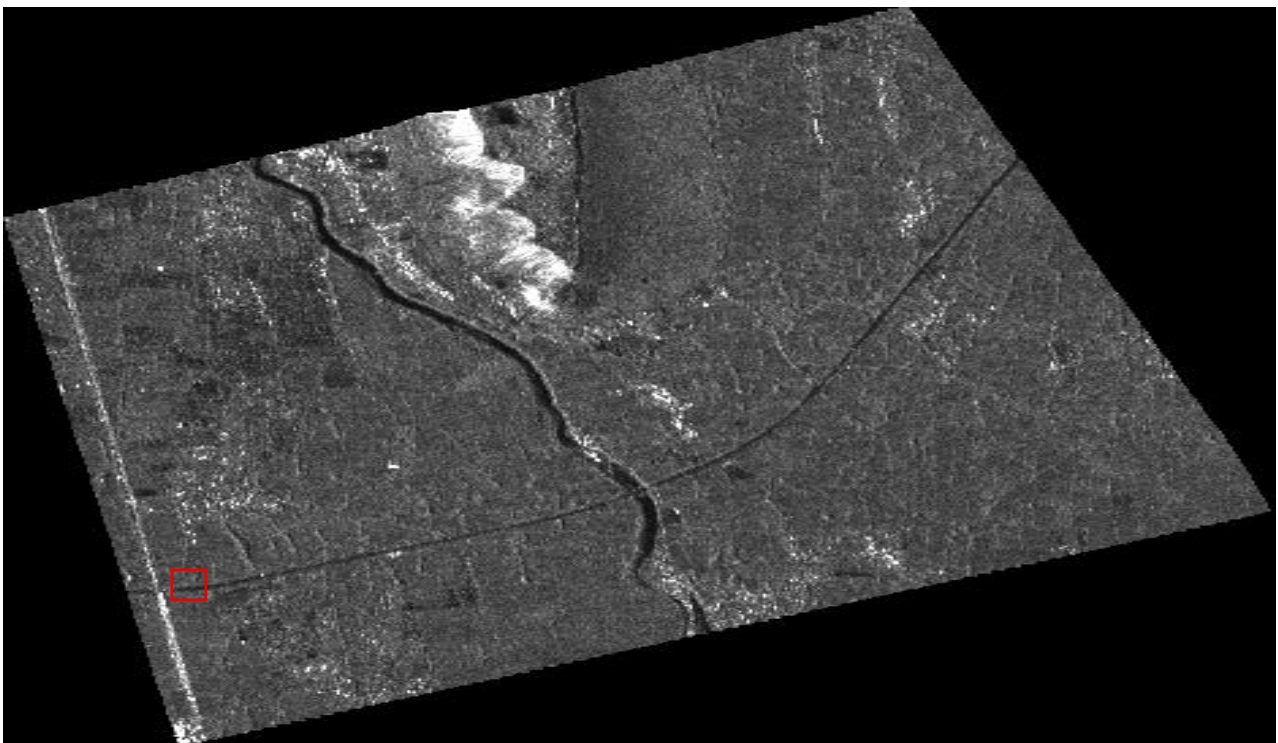
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

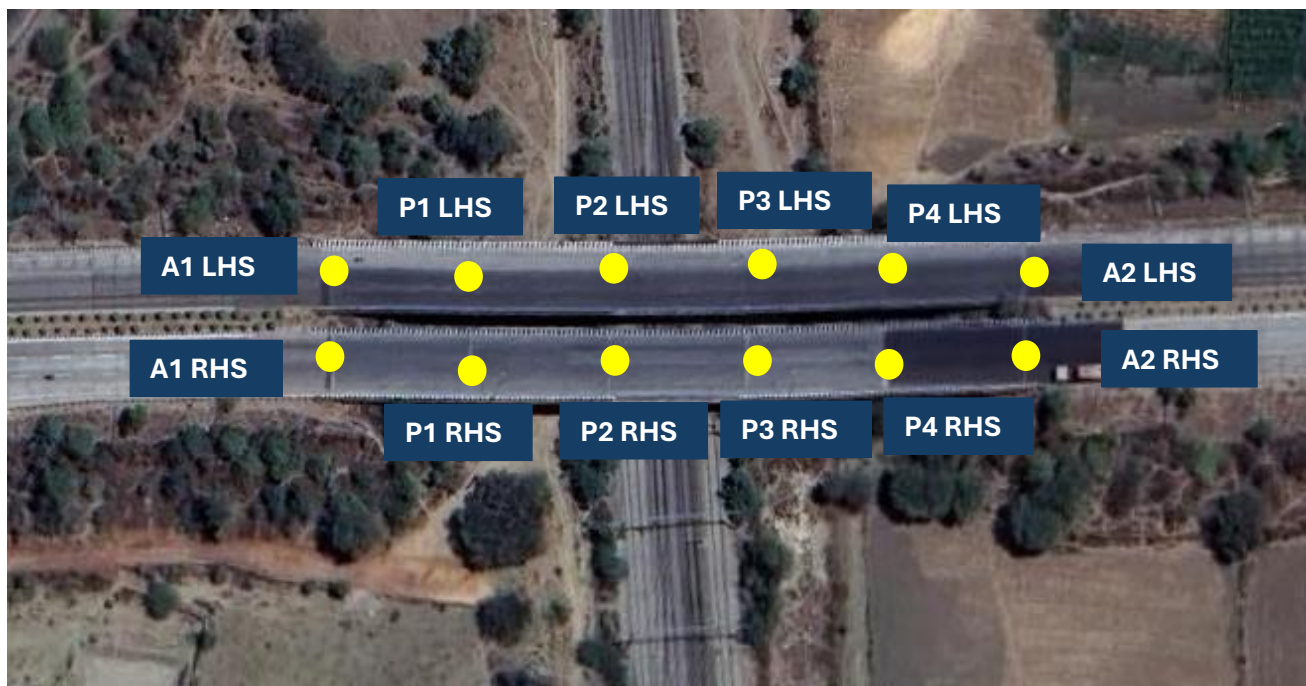


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

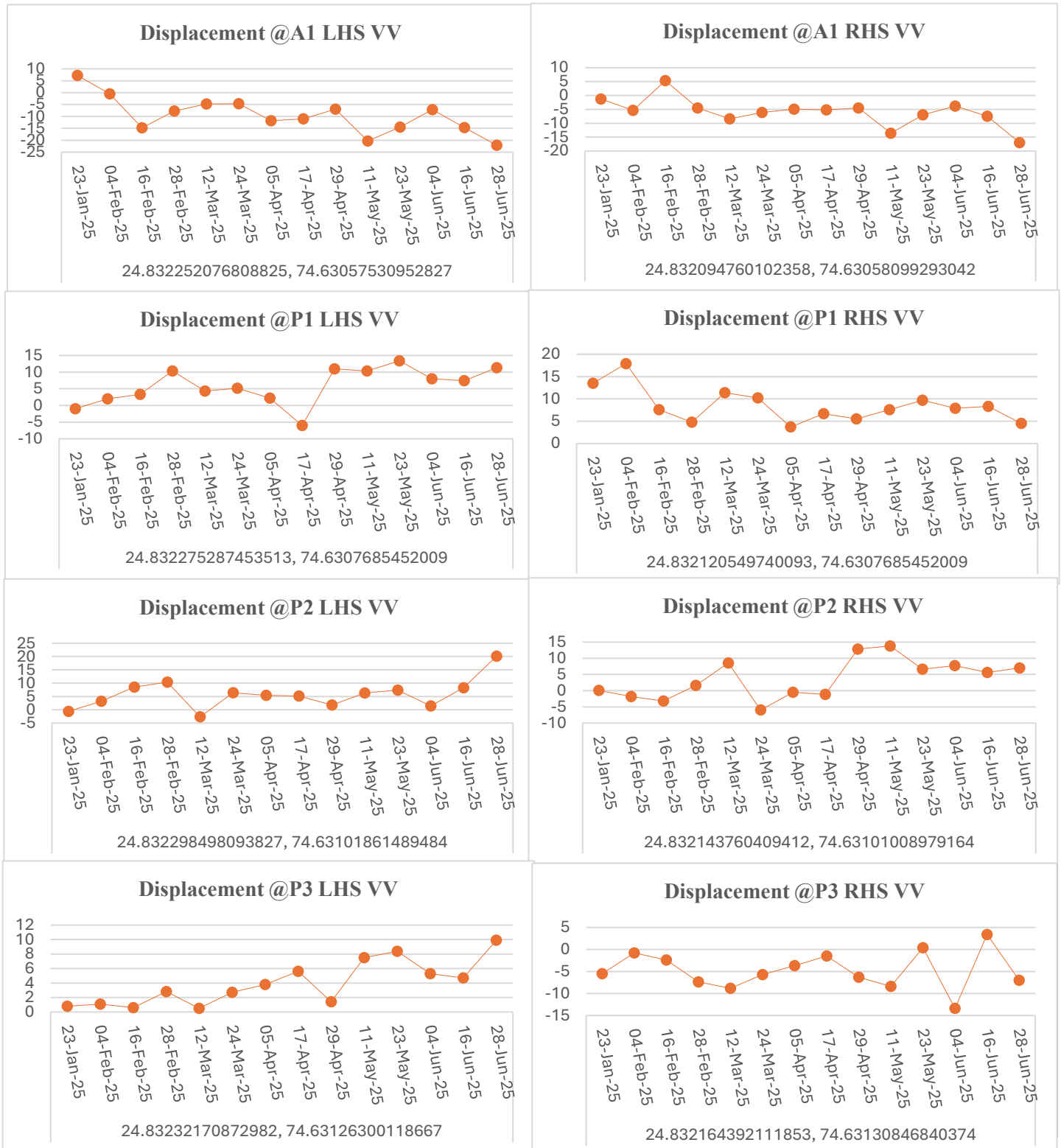
ROB – CH-897+372

Length – 142.21M

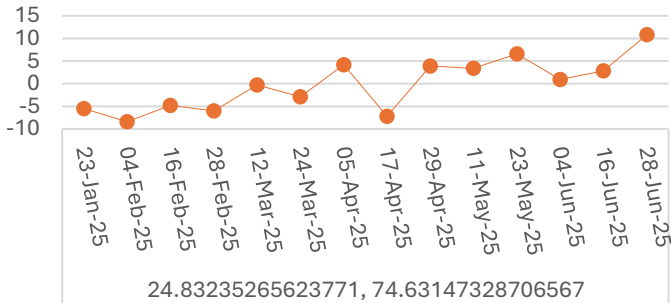


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1	Abutment 1		
P1	Pier 1		
P2	Pier 2		
P3	Pier 3		
P4	Pier 4		
A2	Abutment 2		
LHS	Left Hand Side		
RHS	Right Hand Side		

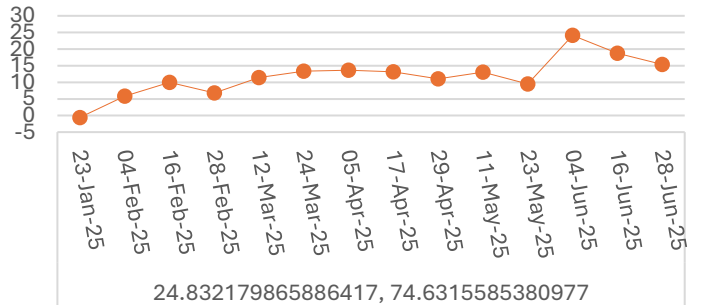
3.1 VV Displacement from 23 Jan 2025 to 28 June 2025



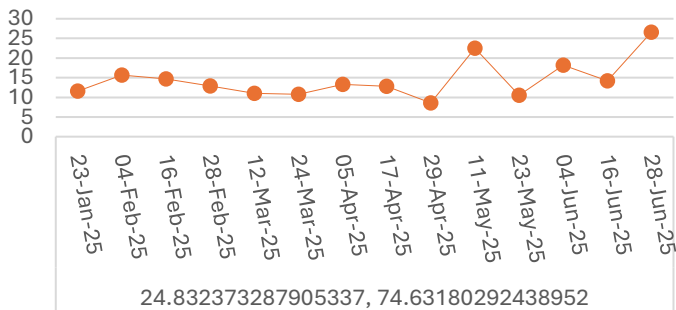
Displacement @P4 LHS VV



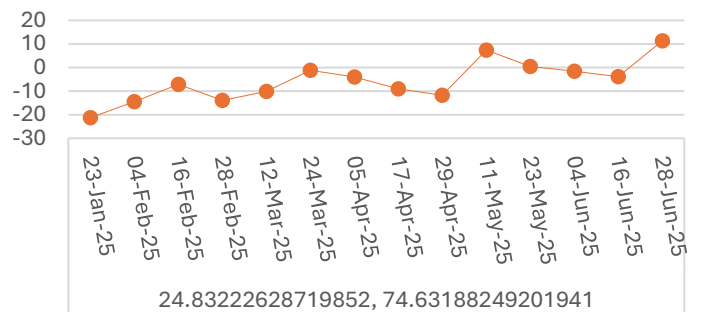
Displacement @P4 RHS VV



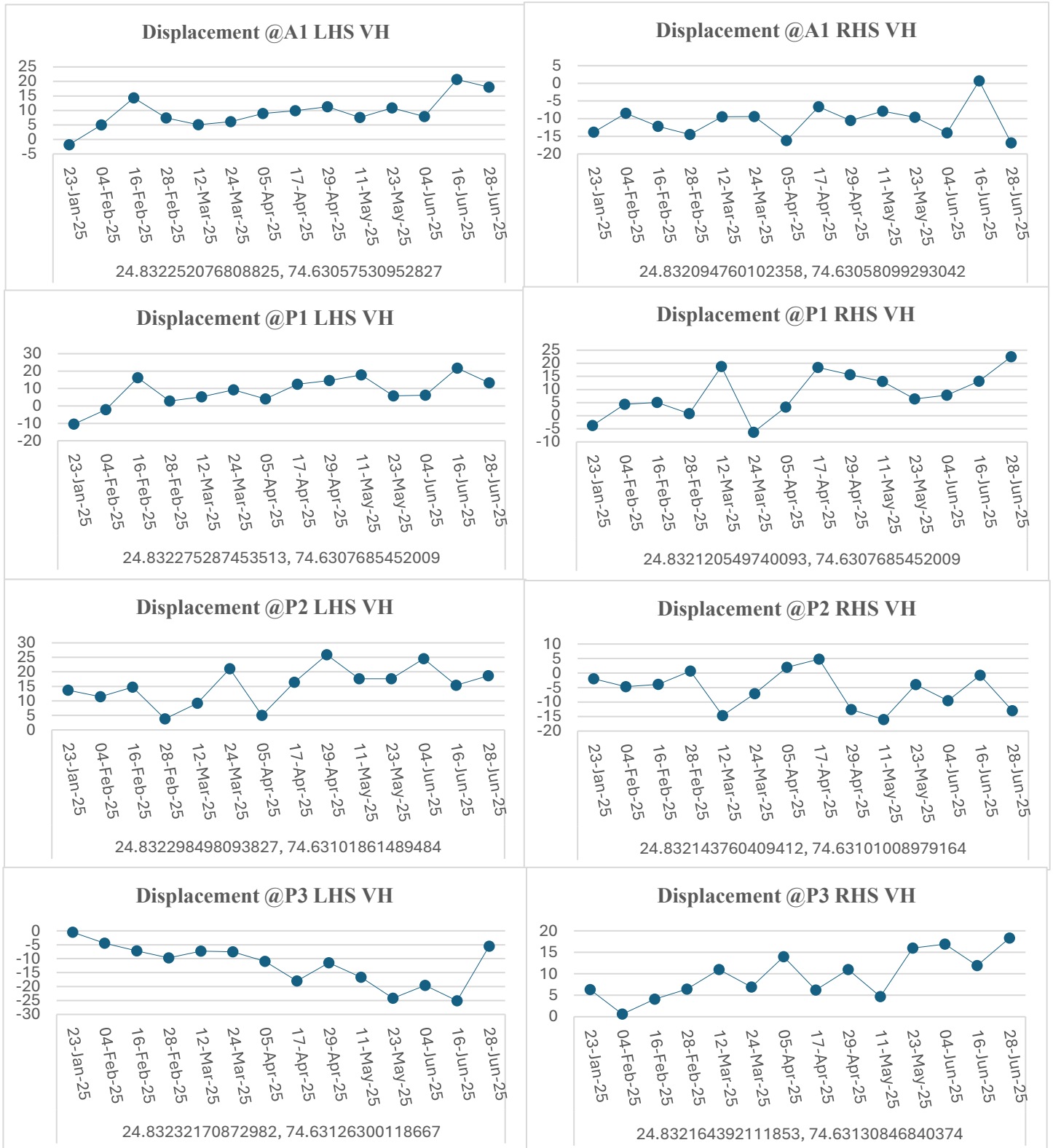
Displacement @A2 LHS VV



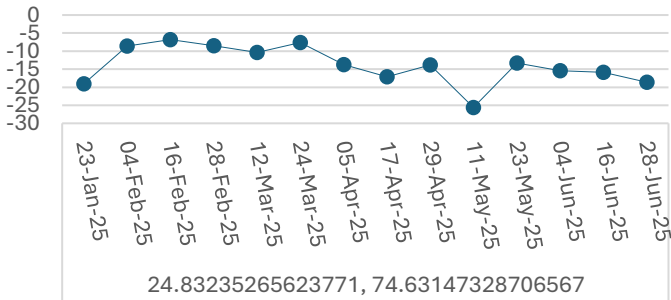
Displacement @A2 RHS VV



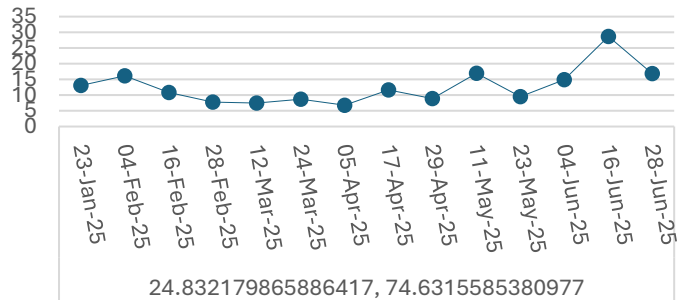
3.2 VH Displacement From 21 June 2024 to 24 March 2025



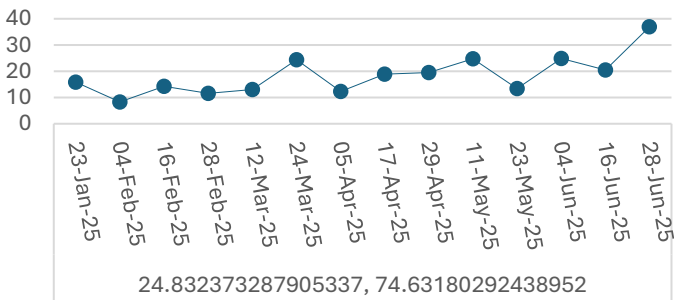
Displacement @P4 LHS VH



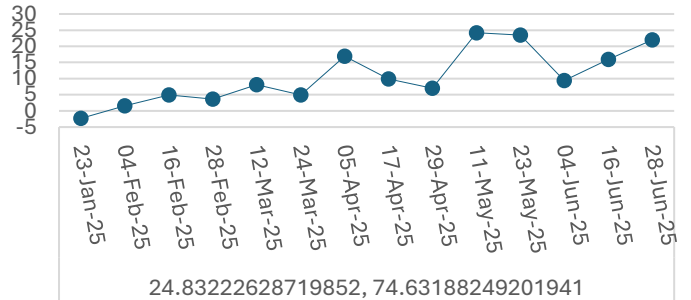
Displacement @P4 RHS VH



Displacement @A2 LHS VH



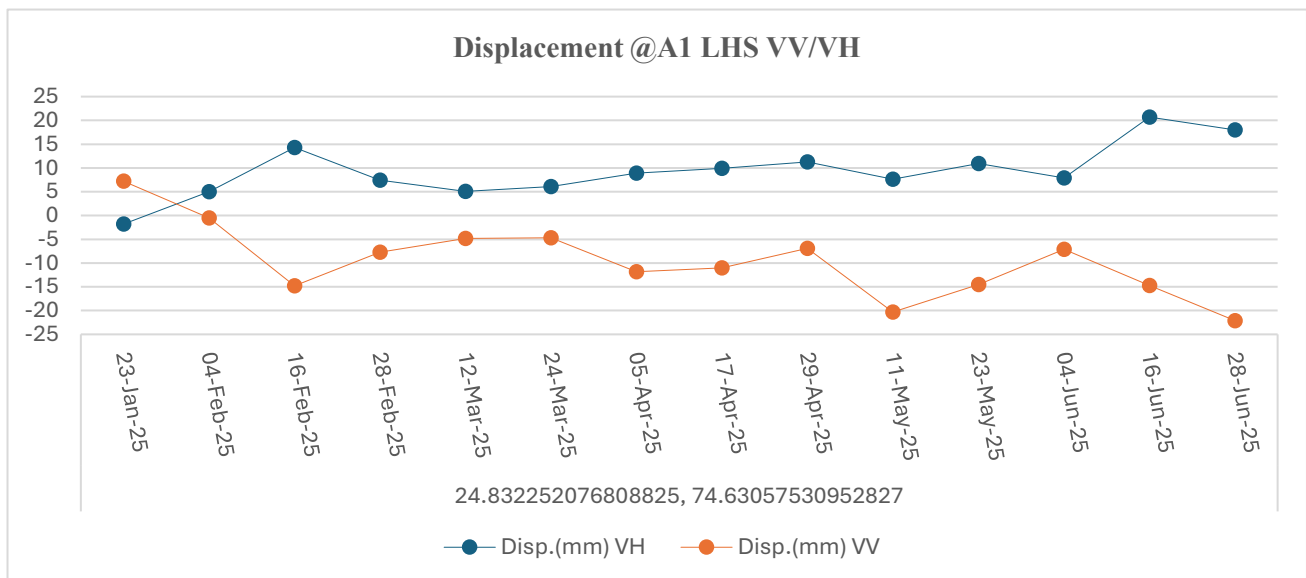
Displacement @A2 RHS VH



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (24.832252076808825, 74.63057530952827)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** shows a net upward (uplift) trend, ranging approximately from –5 mm to +20 mm over the monitoring period.
- **Horizontal displacement (VH)** remains consistently positive, varying approximately from +10 mm to +24 mm, indicating persistent lateral movement in a single dominant direction.

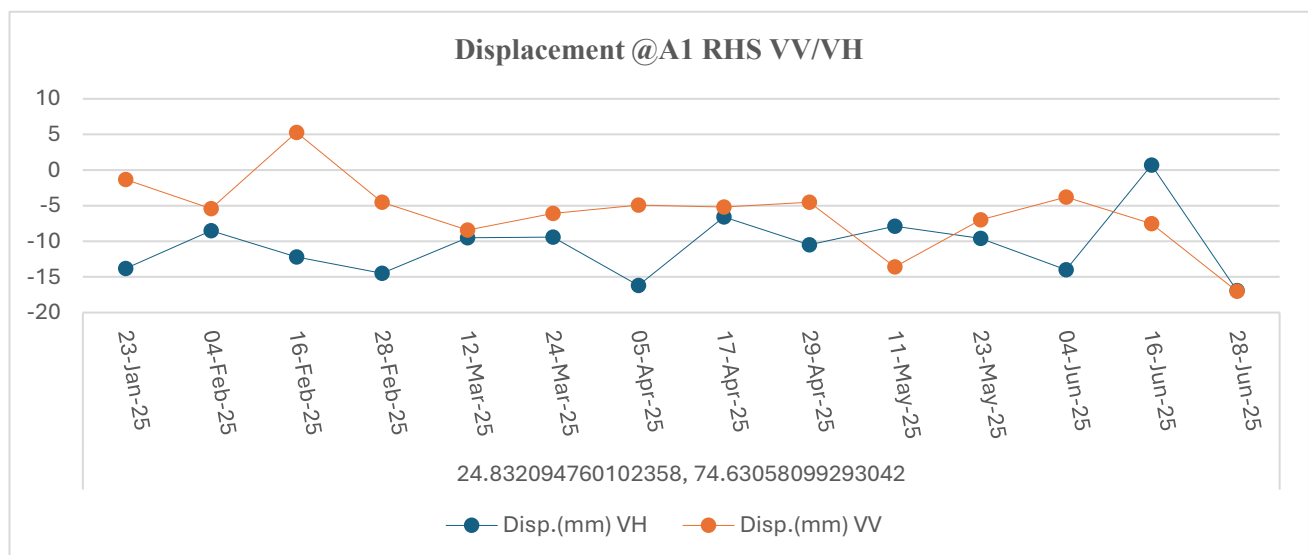
Critical Events & Anomalies:

- **February 2025:** A brief dip in VV (+2 mm) while VH remains positive suggests local vertical adjustment without lateral instability.
- **May 2025:** A pronounced peak in VH accompanied by elevated VV (+12 mm) indicates a combined lateral and vertical response, likely influenced by thermal effects, bearing response, or transient loading conditions.
- **June 2025:** Both VV and VH rise together toward higher values, showing coherent displacement behaviour without abrupt reversals.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (24.832094760102358, 74.63058099293042)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** values are consistently negative, varying approximately between –8 mm and –16 mm, indicating lateral movement in a single dominant direction.
- **Vertical displacement (VV)** values remain predominantly negative, ranging approximately from –2 mm to –17mm, indicating overall settlement-dominated behaviour at A1 RHS.

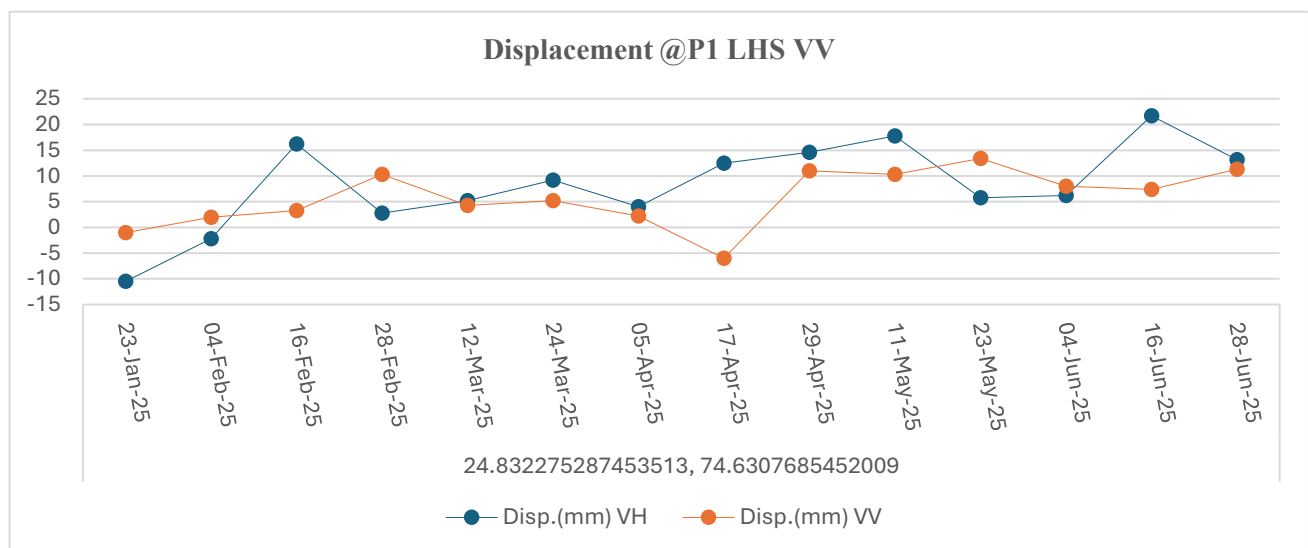
Critical Events & Anomalies:

- **February 2025:** VV reaches a local positive peak (+5 mm) while VH remains negative, indicating short-term vertical rebound without lateral realignment.
- **May 2025:** Simultaneous increase in settlement (VV–14 mm) and negative VH (~–8 mm) suggests a combined vertical–lateral response, possibly linked to thermal effects or bearing restraint behaviour.
- **June 2025:** A sharp VH reduction (–1 mm) followed by immediate reversal indicates a temporary adjustment rather than permanent structural shift.

3.5 SAR Based Analysis at Pier 1 LHS:



P1 LHS (24.832275287453513, 74.6307685452009)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal Displacement (VH)** at P1 LHS shows a generally positive-dominant trend, ranging approximately from -10 mm to +22 mm.
- **Vertical Displacement (VV)** values at P1 LHS range approximately from -6 mm to +14 mm, indicating moderate vertical movement with alternating uplift and settlement phases.

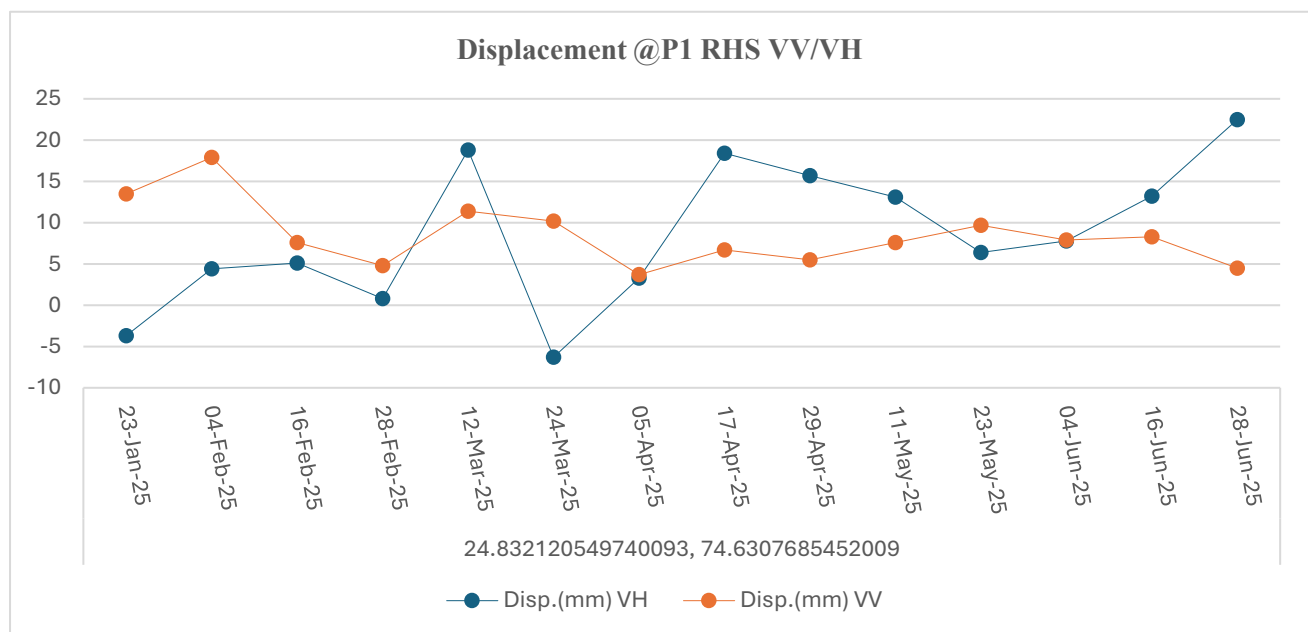
Critical Events & Anomalies:

- **February 2025:** VH shifts from negative to positive values, indicating initial structural adjustment or redistribution of loads at P1 LHS.
- **March 2025:** Concurrent increase in VH with a brief negative VV excursion suggests a structural response
- **June 2025:** VH reaches and maintains higher positive magnitudes, VV remains predominantly positive with reduced variability.

3.6 SAR Based Analysis at Pier 1 RHS:



P1 RHS (24.832120549740093, 74.6307685452009)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal Displacement (VH)** at P1 RHS exhibits a predominantly positive trend, ranging approximately from -7 mm to +22 mm.
- **Vertical Displacement (VV)** Vertical displacement values range approximately from +4 mm to +18 mm.

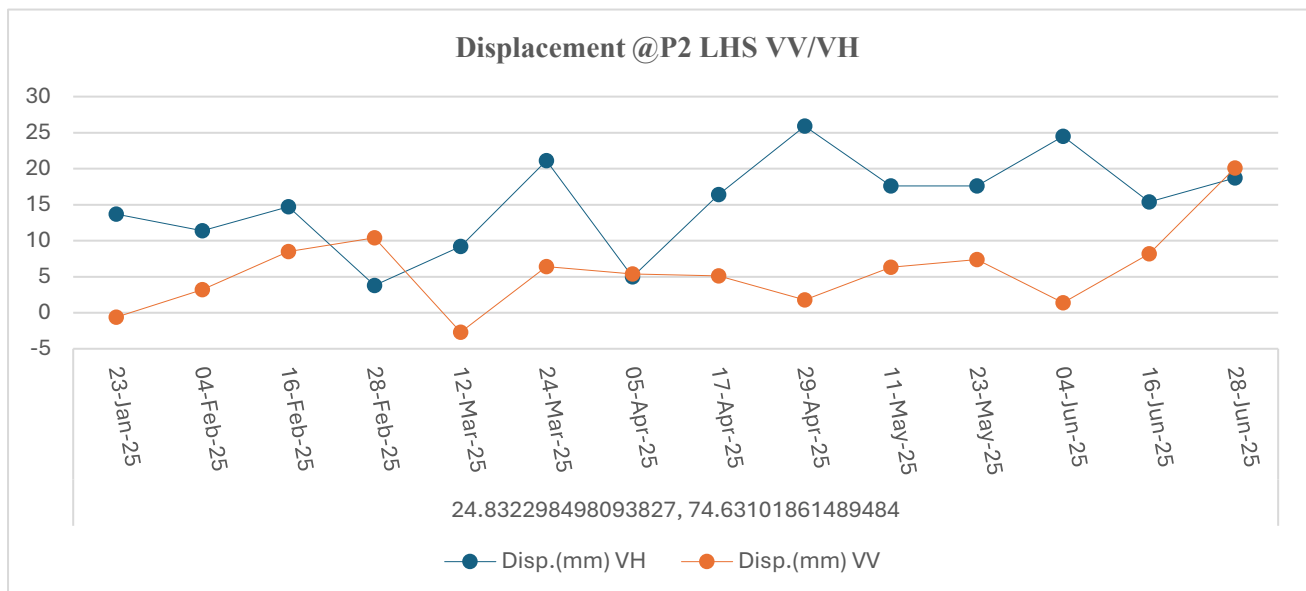
Critical Events & Anomalies:

- **February 2025:** VV reaches an early local peak, while VH transitions from negative to positive, suggesting initial structural adjustment or thermal/load-induced response.
- **April-May 2025:** VH increases steadily to high positive magnitudes, accompanied by stable VV values, reflecting lateral displacement dominance with controlled vertical response.
- **June 2025:** VH attains its maximum positive value, while VV remains moderate and positive, indicating continued lateral movement with no evidence of progressive vertical settlement.

3.7 SAR Based Analysis at Pier 2 LHS:



P2 LHS (24.832298498093827, 74.63101861489484)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal Displacement (VH)** Horizontal displacement at P2 LHS shows a consistently positive trend, ranging approximately from +4 mm to +26 mm.
- **Vertical Displacement (VV)** Vertical displacement values range approximately from –3 mm to +20 mm.

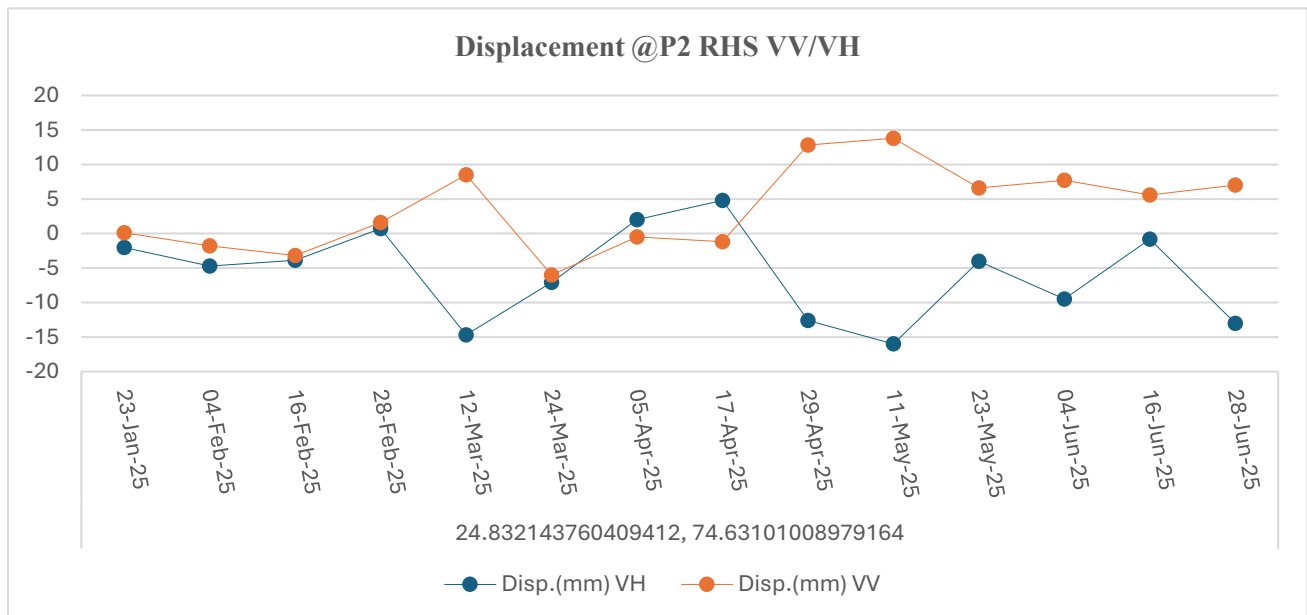
Critical Events & Anomalies:

- **February 2025:** VH displays moderate positive values while VV gradually increases, suggesting initial lateral response accompanied by mild vertical uplift.
- **March 2025:** A pronounced peak in VH (21 mm) coincides with a brief negative VV excursion, indicating a strong structural response possibly due to transient loading or thermal effects.
- **June 2025:** Both VH and VV rise simultaneously, with VV attaining its highest positive magnitude, suggesting combined lateral movement and vertical uplift, potentially linked to seasonal or operational effects

3.8 SAR Based Analysis at Pier 2 RHS:



P2 RHS (24.832143760409412, 74.63101008979164)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal Displacement (VH)** at P2 RHS shows a predominantly negative trend, ranging approximately from -16 mm to +4 mm.
- **Vertical Displacement (VV)** values range approximately from -4 mm to +14 mm, indicating moderate vertical variability with alternating uplift and settlement phases, but no continuous progressive settlement.

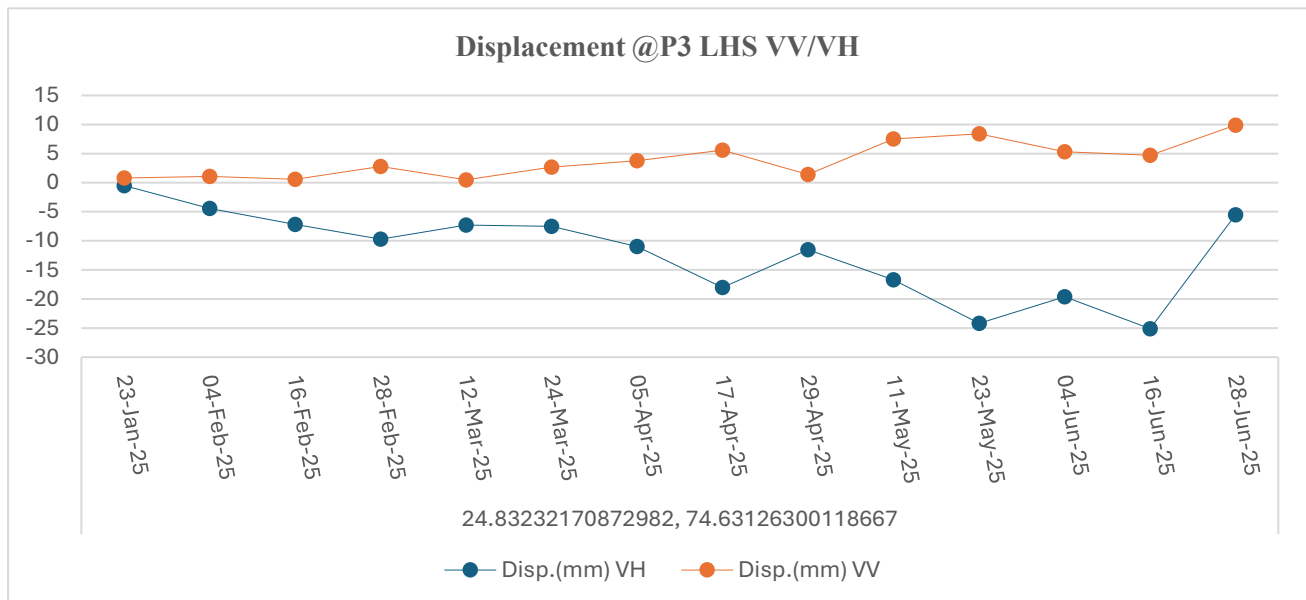
Critical Events & Anomalies:

- **February 2025:** VH remains mildly negative while VV stays close to zero, suggesting initial stable structural behaviour with limited vertical response.
- **April 2025:** VH briefly recovers toward positive values while VV remains slightly negative, reflecting temporary lateral relaxation.
- **June 2025:** VH fluctuations reduce in magnitude while VV stabilizes at moderately positive values, suggesting partial lateral stabilization with sustained vertical uplift.

3.9 SAR Based Analysis at Pier 3 LHS:



P3 LHS (24.83232170872982, 74.63126300118667)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal Displacement (VH)** at P3 LHS shows a consistently negative trend, ranging approximately from –2 mm to –25 mm.
- **Vertical Displacement (VV)** values range approximately from 0 mm to +10 mm, indicating moderate upward movement (uplift) with gradual increase over the monitoring period and limited short-term reversals.

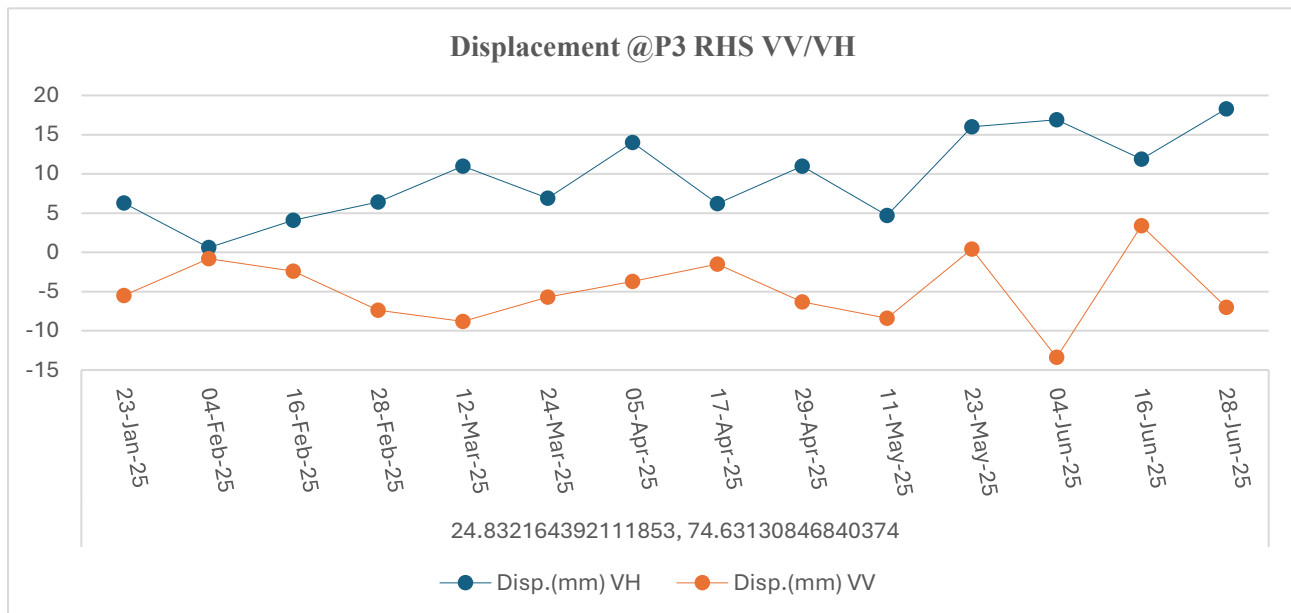
Critical Events & Anomalies:

- **February 2025:** VH transitions from mild negative values to more pronounced negative displacement, suggesting onset of lateral structural movement, while VV remains low and stable.
- **May 2025:** VH records its maximum negative displacement (–25 mm), while VV peaks around +8 mm, highlighting opposing displacement trends and potential differential response at this location
- **June 2025:** VH shows partial recovery toward –5 mm, while VV attains its highest positive value, suggesting reduced lateral instability with continued vertical uplift

3.10 SAR Based Analysis at Pier 3 RHS:



P3 RHS (24.832164392111853, 74.63130846840374)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal Displacement (VH)** Horizontal displacement at P3 RHS exhibits a predominantly positive trend, ranging approximately from 0 mm to +18 mm.
- **Vertical Displacement (VV)** values range approximately from -14 mm to +4 mm, indicating moderate vertical movement dominated by settlement phases, with occasional short-term uplift events.

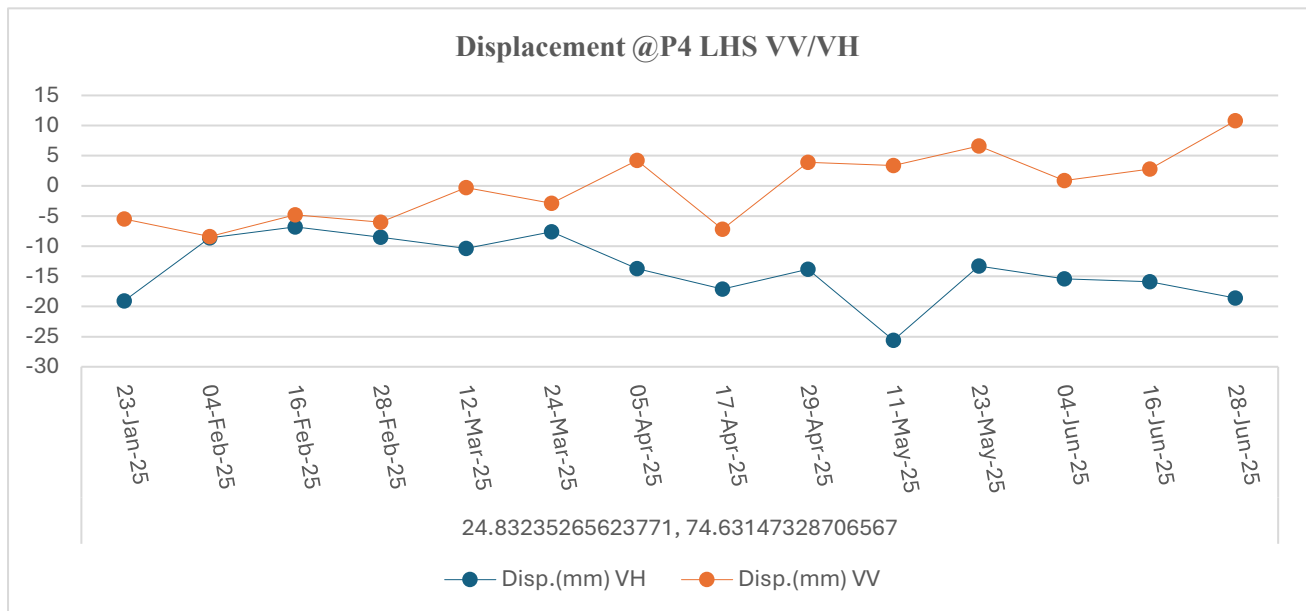
Critical Events & Anomalies:

- **February 2025:** VH shows an initial reduction toward near-zero values before recovering, while VV remains mildly negative, suggesting early-stage adjustment without significant vertical distress.
- **May 2025:** VH increases sharply to around +15 mm, while VV remains negative, highlighting amplified lateral displacement with continued settlement.
- **June 2025:** VH attains its highest positive value (+18 mm), while VV returns to negative levels, indicating persistent lateral displacement with limited vertical recovery.

3.11 SAR Based Analysis at Pier 4 LHS:



P4 LHS (24.83235265623771, 74.63147328706567)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal Displacement (VH)** Horizontal displacement shows high variability with strong negative dominance, ranging roughly between -6 mm and -26 mm.
- **Vertical Displacement (VV)** Vertical displacement exhibits moderate variability with a gradual transition from early settlement to dominant uplift over the monitoring period.

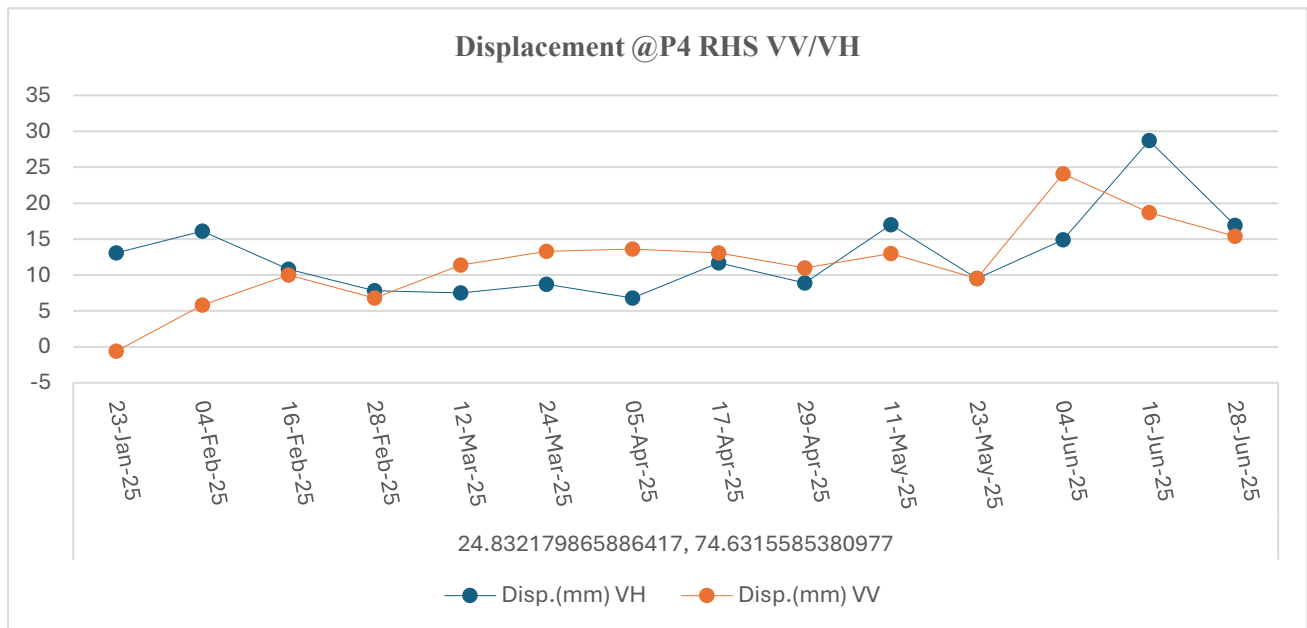
Critical Events & Anomalies:

- **January 2025:** VH starts near -18 mm, while VV is moderately negative (-5 mm), indicating early coupled lateral and vertical settlement.
- **April 2025:** VV rises sharply to $+4$ mm, while VH deteriorates to -14 mm, indicating a decoupled response with vertical uplift occurring alongside lateral settlement.
- **June 2025:** VV peaks at $+11$ mm, while VH remains strongly negative (-18 mm), confirming persistent lateral settlement combined with increasing vertical uplift toward the end of the monitoring period.

3.12 SAR Based Analysis at Pier 4 RHS:



P4 RHS (24.832179865886417, 74.6315585380977)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal Displacement (VH)** Horizontal displacement ranges approximately between +10 mm and –12 mm, showing moderate variability with an overall declining tendency across the monitoring period.
- **Vertical Displacement (VV)** Vertical displacement varies from about –5 mm to –22 mm, demonstrating a predominantly downward (settlement) trend. Unlike VH, VV shows limited recovery once negative displacement increases, suggesting progressive vertical movement.

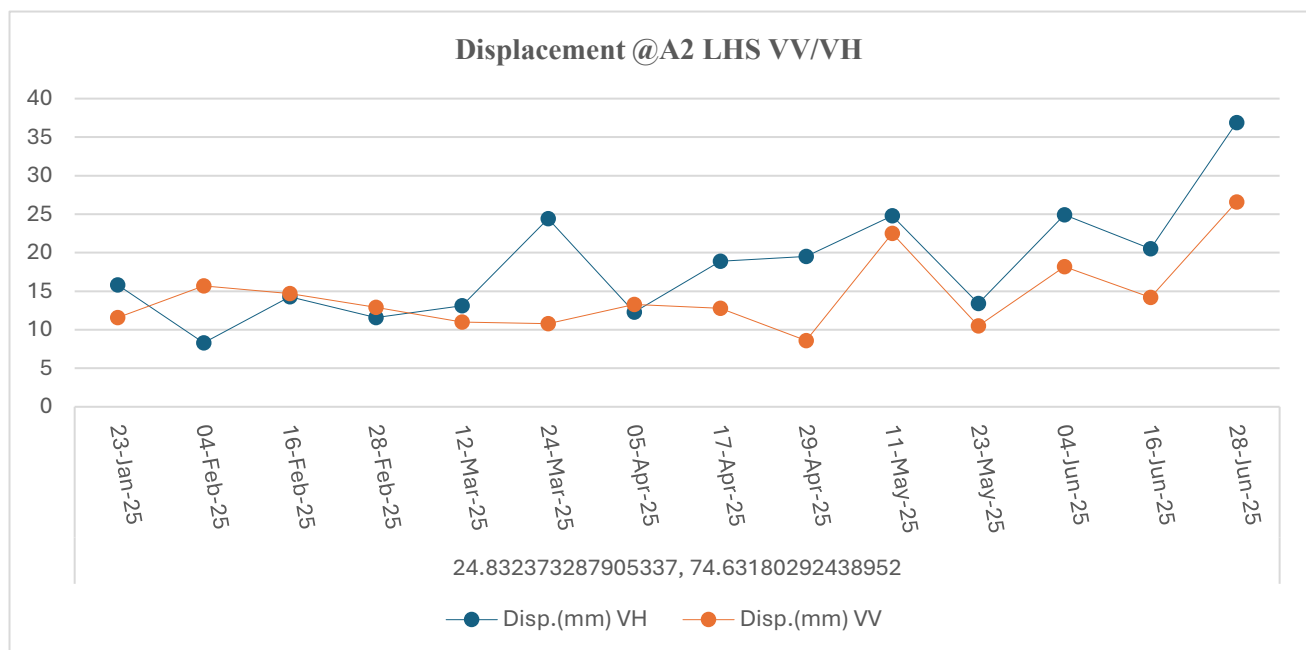
Critical Events & Anomalies:

- **January 2025:** VH records a clear positive peak (+10 mm) while VV is mildly negative (–5 mm), indicating an initial lateral response with minimal vertical settlement.
- **April 2025:** A pronounced simultaneous drop occurs, with VH reaching around –13 mm and VV near –15 mm, indicating a combined lateral and vertical displacement event
- **June 2025:** The monitoring period concludes with sustained negative displacements, with VH around –12 mm and VV reaching its maximum settlement.

3.15 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (24.832373287905337, 74.63180292438952)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal Displacement (VH)** Horizontal displacement shows high variability with strong positive dominance, ranging roughly between +8 mm and +36 mm.
- **Vertical Displacement (VV)** Vertical displacement exhibits moderate to high variability with an overall positive trend over the monitoring period.

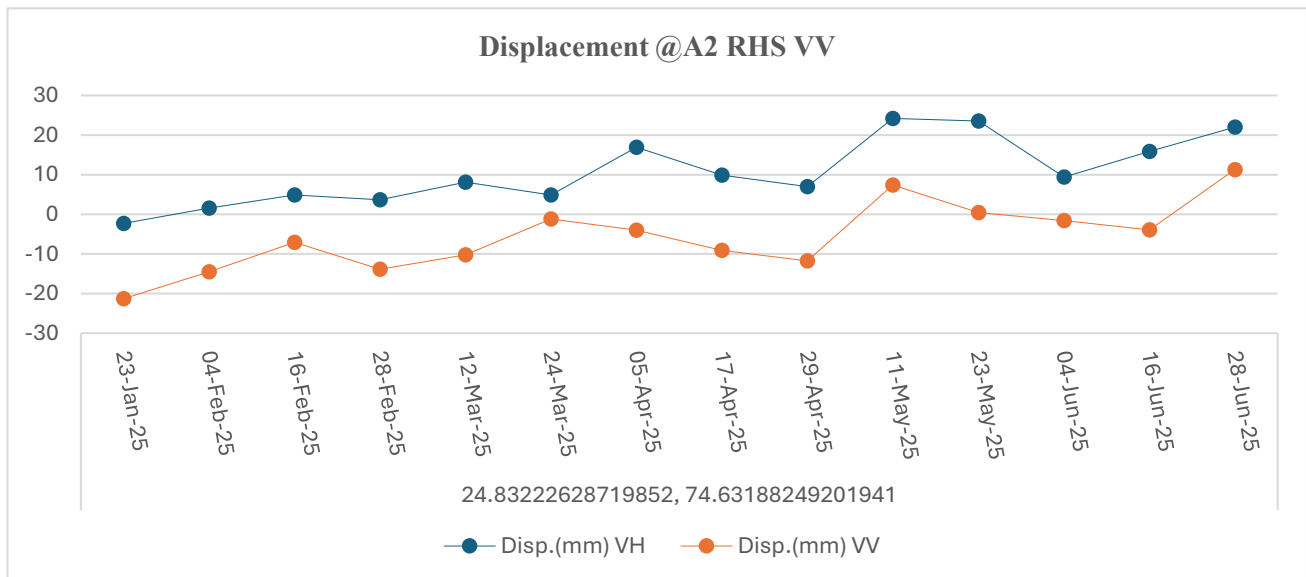
Critical Events & Anomalies:

- **January 2025:** VH begins around +15 mm, VV is near +12 mm, indicating early coupled lateral, vertical uplift.
- **May 2025:** Both VH and VV peak simultaneously (VH +25 mm, VV +22 mm), representing a strong combined lateral and vertical uplift event.
- **June 2025:** VH rises again to +25 mm, while VV increases to +18 mm, indicating renewed deformation activity.

3.16 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (24.83222628719852, 74.63188249201941)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal Displacement (VH)** Horizontal displacement exhibits moderate to high variability with a strong positive dominance, ranging roughly between -2 mm and $+24$ mm.
- **Vertical Displacement (VV)** Vertical displacement shows moderate variability with an overall improving trend over the monitoring period. VV values range approximately from -22 mm to $+12$ mm.

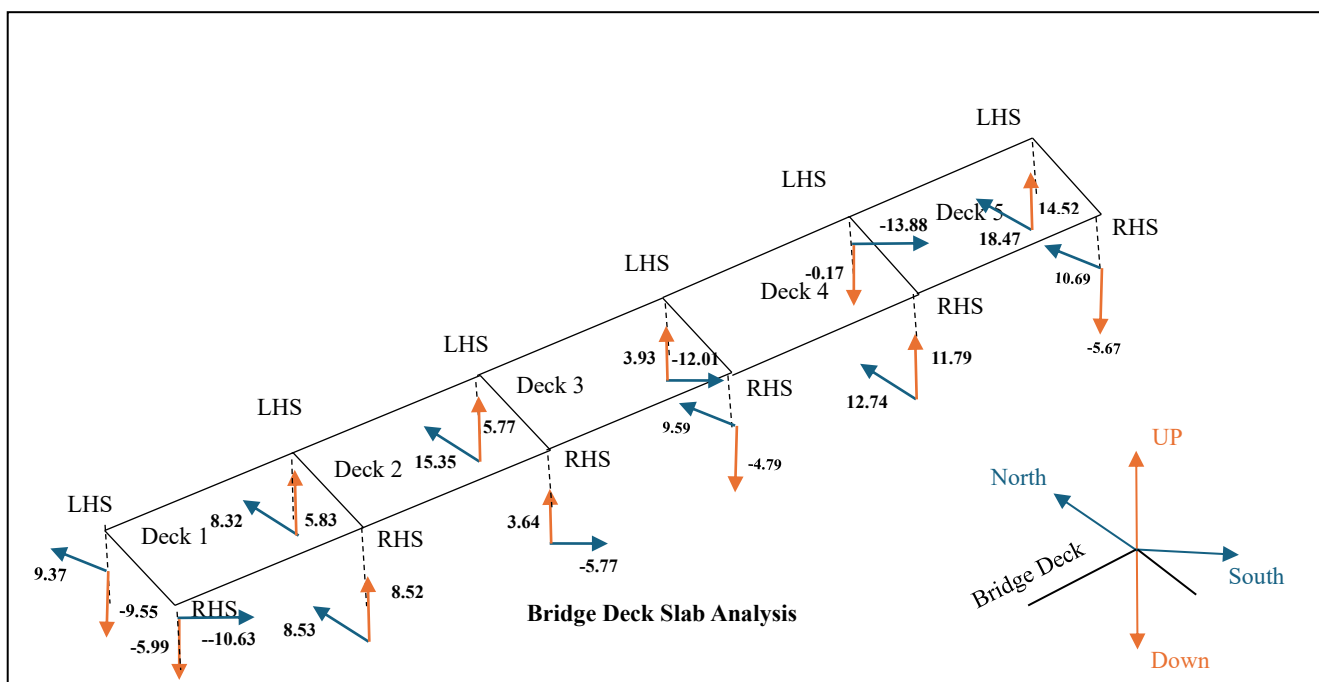
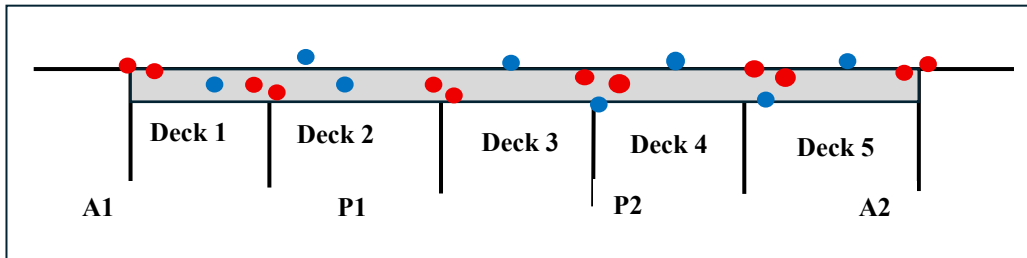
Critical Events & Anomalies:

- **February 2025:** VH improves to $+1$ mm, while VV recovers to -15 mm, marking a partial reduction in settlement severity
- **April 2025:** VH increases sharply to $+17$ mm, while VV drops slightly to -5 mm, highlighting a lateral-dominant deformation phase.
- **June 2025:** VH reduces slightly to $+10$ – 16 mm, while VV increases to $+12$ mm, confirming sustained vertical uplift with continued lateral displacement toward the end of the monitoring period.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- The bridge deck slab exhibits pronounced non-uniform vertical displacement along its longitudinal extent, indicating differential behaviour across Decks 1 to 5. Localised downward displacements are observed at several RHS locations, notably Deck 1 (−9.6 mm), Deck 3 (−12.0 mm), Deck 4 (−0.2 mm), and Deck 5 (−5.7 mm), highlighting intermittent settlement behaviour predominantly on the RHS.

Horizontal Displacement Behaviour:

- Horizontal displacement shows strong directional variability and asymmetry across all decks, reflecting combined translational and torsional effects.

CONCLUSION:

- The bridge deck slab demonstrates a highly non-symmetric deformation pattern, with both vertical uplift and horizontal displacement increasing toward the higher-numbered decks. Overall, the observed behaviour highlights critical deformation zones at Decks 4 and 5, warranting focused monitoring and correlation with structural detailing and boundary conditions.